

Data gathering and solar wind plotting

GOALS

1. The goal of this exercise is to gather data from the HEIMDALL campaign and identify a research questions that is relevant to the field of space physics.
2. Being able to extract and collect in-situ observations of the solar wind.

Task

This exercise is divided in three parts. The first one consists of gathering and combining figures/information obtained during the campaign in a single presentation (or document), and briefly describing the observations. This should give an overview of the entire event and help identify a research topic. The second part consists of learning how to extract and plot solar wind data for the event. The third part consists of identifying interval(s) of interest and formulating a scientific question.

1. Part 1: Data gathering:

Figures and data from the campaign have been placed in "files" on CANVAS. Download the figures and create a document (or presentation) in which you combine the observations. After this is done, write a few sentences describing what physical quantity(ies) are measured by each of the instruments and describe the observations. Note that you can also plot the Eiscat data yourself instead of using the pre-made figures. It could also be interesting to plot TEC maps for the event.

2. Part 2: Solar wind data extracting and plotting

The goal is to learn how to access solar wind data, especially the magnetic field, proton density, and velocity. Data from the ACE satellite [Stone et al., 1998] available at <http://cdaweb.gsfc.nasa.gov/cdaweb/istp-public/>, for instance.

- (a) Go to the webpage above-mentioned, choose "ACE", and click on "submit" at the bottom of the page.
- (b) If not already "cleared", CLEAR all checkboxes and then select:
 - i. "AC_Ko_MFI: ACE Magnetic Field 5-Minute Key Parameters [PRELIM] - N. Ness (Bartol Research Institute)"
 - ii. "AC_Ko_SWE: Ko - ACE Solar Wind Experiment 5-Minute Key Parameters [PRELIM] - D. J. McComas (Southwest Research Institute)"

- (c) Select “submit” at the bottom of the page. (Note that some of the data with other resolutions (and not “preliminary” will become available at a later stage).
- (d) Choose the starting time 2026/02/02 12:00:00.000 and a Stop time of 2026/02/02 23:59:00.000. Then select the parameters of interest.
- “[PRELIMINARY VALUES - BROWSE USE ONLY] B-field magnitude”
 - “[PRELIMINARY VALUES - BROWSE USE ONLY] Magnetic Field, Cartesian GSE coord”
 - “[PRELIMINARY VALUES - BROWSE USE ONLY] Solar Wind Proton Number Density, scalar”
- PRELIMINARY VALUES - BROWSE USE ONLY Solar Wind Bulk Speed
- (e) You can then select “Plot Data” to create quicklooks and create files to download (either ASCII/CSV, or CDF). Choose your favorite data format to download and press “submit”.
- (f) Calculate a delay time between the observations made by ACE and their arrival at Earth.
- (g) Include the figures of the solar wind you created in the document (or presentation) gathering all the observations (from part 1).

3. Part 3: Scientific questions

Scrutinize the observations and identify “events” of interest. Create a ~5 minute presentation highlighting what you think are the most important observations and what you think was most interesting. Finally, think about what would you like to study in more detail as a part of your project?

Try thinking about some of the following questions :

- Why did the electron density peak around midnight?
- What are the tiger stripes-like structures observed in velocity, density, and temperature in the VHF?
- Why is there a hot ion layer in the VHF data?
- What is the hot-cold T_e boundary in the VHF?
- Was there strong precipitation that caused riometer absorption?
- Did we observe substorm(s) and do the different instruments (e.g., radars, magnetometers, camera) show consistent features?

We will discuss your ideas on Thursday 19.02.2026. (At this point, you do not need to know all the answers)